

Table S3. Classical molecular-model settings. Each fingerprint was used in a separate endpoint-specific model. Shared settings are listed once; braces indicate hyperparameter search spaces. Model selection used validation AUPRC. These frozen molecular-model settings are unchanged by the revision to transcriptomic fitting. Shared settings: training seed: 1.

Representation	Size	Model	Selection / tune	Class weighting	Settings
MACCS structural	167.0	Logistic regression	C in {0.001, 0.01, 0.1, 1, 10, 100}; selected by validation AUPRC	class_weight=balanced	max_iter=2000
Morgan fingerprint	2048.0	Logistic regression	radius=2; C in {0.001, 0.01, 0.1, 1, 10, 100}; selected by validation AUPRC	class_weight=balanced	max_iter=2000
RDKit topological fingerprint	2048.0	Logistic regression	C in {0.001, 0.01, 0.1, 1, 10, 100}; selected by validation AUPRC	class_weight=balanced	max_iter=2000
MACCS structural	167.0	XGBoost	max_depth in {3, 4, 6}; learning_rate in {0.05, 0.1}; selected by validation AUPRC	scale_pos_weight=1	n_estimators=300; subsample=0.8; colsample_bytree=0.8; early_stopping_rounds=10; eval_metric=auprc; n_jobs=1
Morgan fingerprint	2048.0	XGBoost	radius=2; max_depth in {3, 4, 6}; learning_rate in {0.05, 0.1}; selected by validation AUPRC	scale_pos_weight=1	n_estimators=300; subsample=0.8; colsample_bytree=0.8; early_stopping_rounds=10; eval_metric=auprc; n_jobs=1
RDKit topological fingerprint	2048.0	XGBoost	max_depth in {3, 4, 6}; learning_rate in {0.05, 0.1}; selected by validation AUPRC	scale_pos_weight=1	n_estimators=300; subsample=0.8; colsample_bytree=0.8; early_stopping_rounds=10; eval_metric=auprc; n_jobs=1